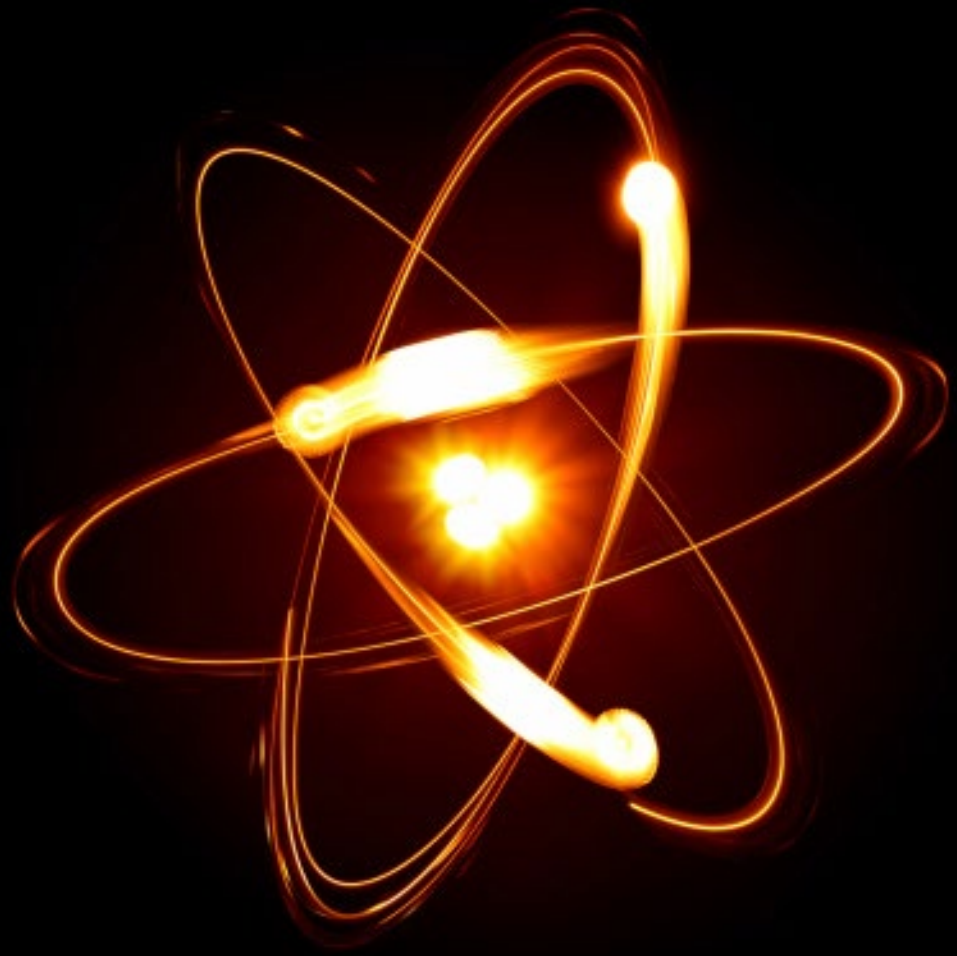
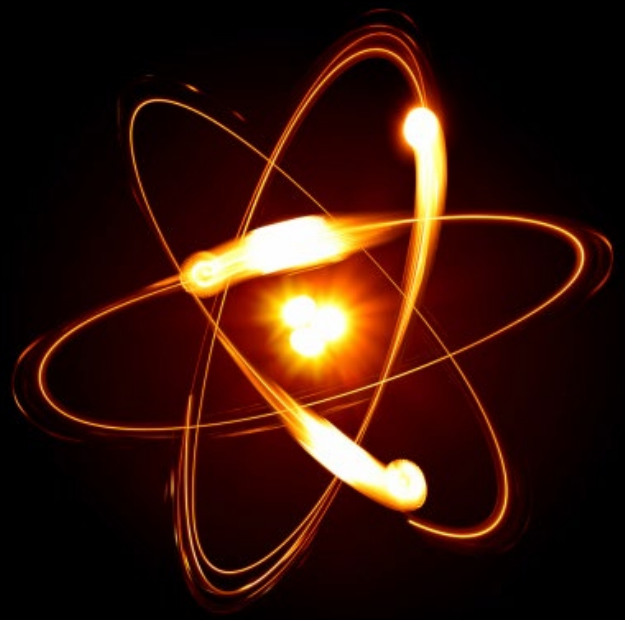
Chapter 5.

Bohr's Model of the Atom



Beatriz González del Río

Quantum Physics



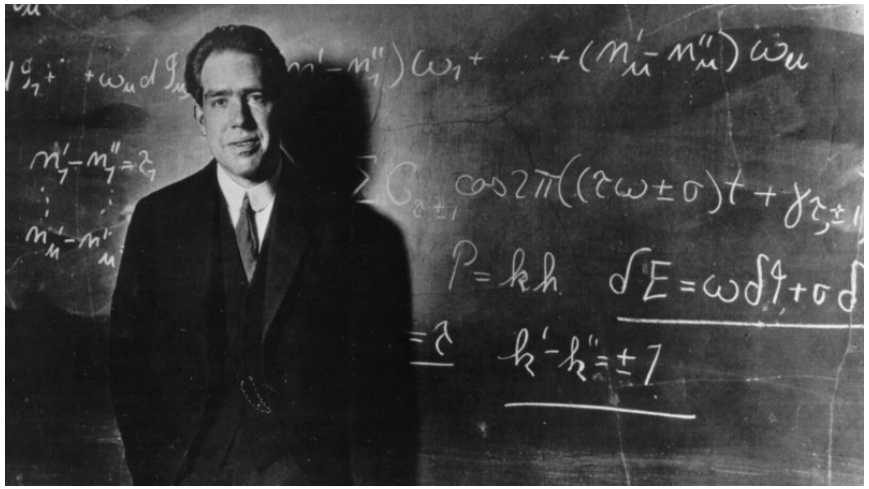
Chapter 5.

Bohr's Model of the Atom

- Bohr's postulates
- Bohr's model: total energy, emission frequencies
- Correction for finite nuclear mass
- Atomic energy states: Franck-Hertz experiment
- The correspondence principle
- A critique to the old quantum theory

Bohr's postulates

First atomic model to provide an explanation for some atomic spectra



1. An electron in an atom moves in a circular orbit about the nucleus under the influence of the Coulomb attraction between the electron and the nucleus, **obeying the laws of classical mechanics**.

2. The electron moves in an orbit for which its orbital angular momentum L is an integer multiple of $\hbar$.

3. Despite the fact that it is constantly accelerating, an electron moving in such an allowed orbit does NOT radiate electromagnetic energy. Therefore, its total energy E remains constant.

4. Electromagnetic radiation is emitted if an electron, initially moving in an orbit of total energy E_i , discontinuously changes its motion so that it moves in an orbit of total energy E_f . The frequency of the emitted radiation υ is equal to the quantity $(E_i - E_f)/h$.

Which of Bohr's postulates are classical and which are genuinely new quantum assumptions?

Bohr's postulates

Meaning of the postulates?

1. The first assumes the existence of the nucleus
2. The second introduces quantization.
3. The third removes the problem of stability. It is based on the experimental fact that atoms are stable.
4. The fourth is based on Einstein's expression for the frequency of a photon.

The postulates are a mixture of classical concepts with the new ideas

Bohr's model

Consider an atom with a nucleus of charge $+Ze$ and an electron revolving about the nucleus in a circular orbit of radius r . We suppose that the nucleus remains fixed in space (the nucleus has an infinite mass). The condition of mechanical stability of the electron is:

$$\frac{Ze^2}{4\pi\epsilon_0 r^2} = m_e \frac{v^2}{r} \qquad (1)$$

Moreover, we have for the orbital angular momentum of the electron

$$\vec{L} = \vec{r} \times \vec{p} \quad \Longrightarrow \quad m_e v r = L$$

Applying the quantization condition for L, we have: $L = m_e v r = n\hbar$

Therefore $v = \frac{n\hbar}{m_e r}$ and substituting into (1)

$$\frac{Ze^2}{4\pi\epsilon_0 r^2} = m_e \frac{n^2\hbar^2}{m_e^2 r^3}$$

AND .....

Bohr's model

$$r = 4\pi\epsilon_0 \frac{\hbar^2 n^2}{Ze^2 m_e}$$

The circular orbits are quantized, i.e. they are restricted to those whose radii are as given above

$$v = \frac{n\hbar}{m_e r} = \frac{1}{4\pi\epsilon_0}\frac{Ze^2}{n\hbar} \qquad n = 1, 2, 3\ldots$$

What about r?

Consider a hydrogen's atom, $Z = 1$, $n = 1$

$r = 0.529 \cdot 10^{-10} m \cong 0.529$ Å radius of hydrogen's atom

$v = 2.3 \cdot 10^6 \frac{m}{s} \ll c$ (*for small Z*)

Bohr's model: Total energy

We define the potential energy to be zero when the electron is infinitely distant from the nucleus. Then the potential energy V at a distance r is

$$V(r) = -\int_r^\infty \frac{Ze^2}{4\pi\epsilon_0 r'^2} dr' = -\frac{Ze^2}{4\pi\epsilon_0 r}$$

which is negative.

The kinetic energy of the electron, K, is

$$K = \frac{1}{2} m_e v^2 = \frac{1}{2} \frac{1}{4\pi\epsilon_0} \frac{Ze^2}{r}$$

(We have taken into account the condition of mechanical stability)

Bohr's model: Total energy

The total energy is:

$$E_{total} = K + V = \frac{1}{2}\frac{Ze^2}{4\pi\epsilon_0 r} - \frac{Ze^2}{4\pi\epsilon_0 r} = -\frac{1}{2}\frac{Ze^2}{4\pi\epsilon_0 r}$$

which is obviously negative.

Substituting the expression for *r*

$$E_{total} = -\frac{Z^2 e^4 m_e}{2\hbar^2 (4\pi\epsilon_0)^2}\frac{1}{n^2} \qquad n = 1, 2, 3\ldots$$

The total energy is quantized. The lowest energy state corresponds to n=1 (this is the most stable state).

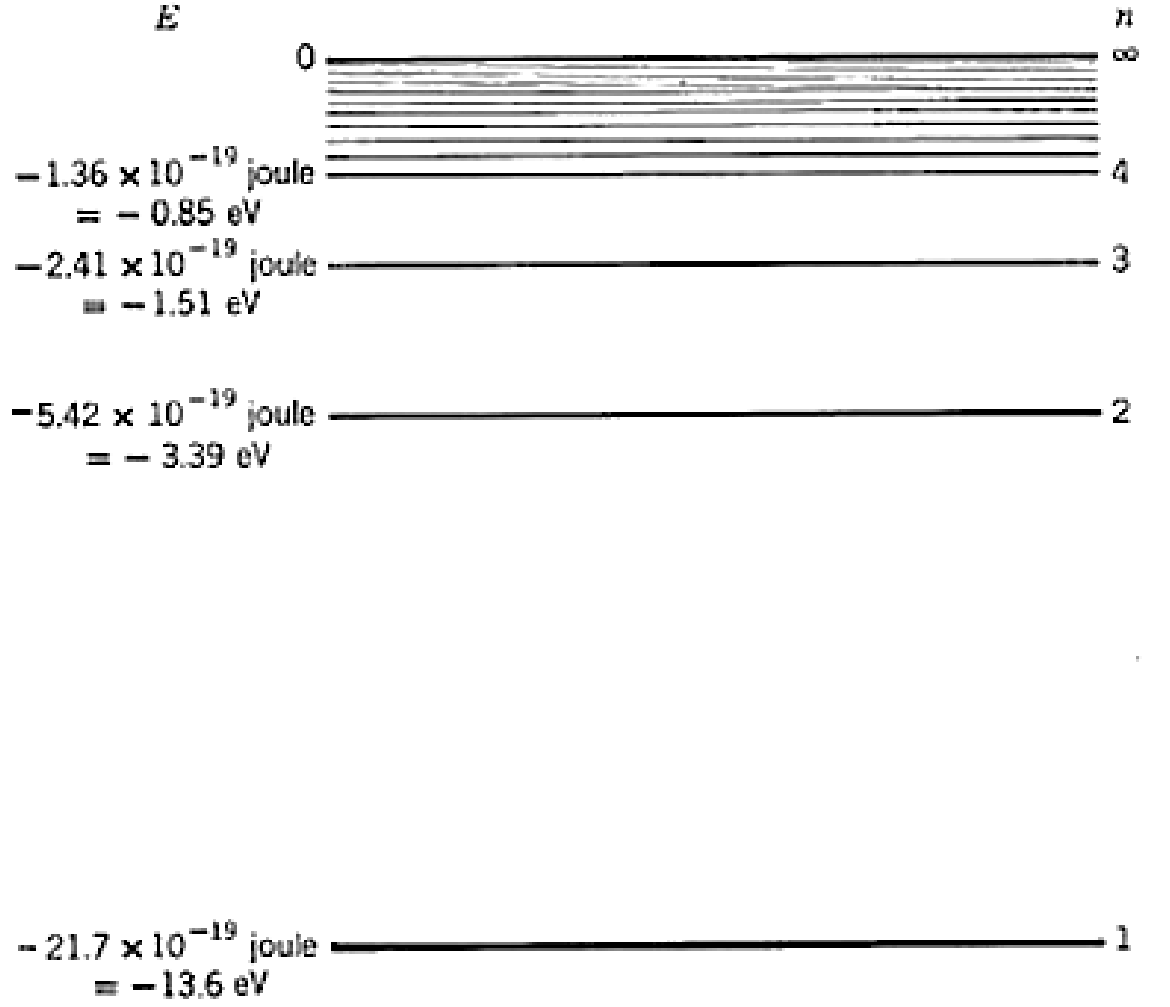


Figure 4-11 An energy-level diagram for the hydrogen atom.

Bohr's model: Emission frequencies

According to the 4th postulate, when an electron jumps from one orbit n_i (quantum state) to another n_f, then it emits radiation with a frequency ν, such that : (in terms of the reciprocal wavelength κ)

$$\kappa = \frac{1}{\lambda} = \frac{\nu}{c} = \frac{1}{c}\frac{E_i - E_f}{h} = \frac{Z^2 e^4 m_e}{4(4\pi\epsilon_0)^2\hbar^3\pi c}\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) = R_\infty Z^2\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right)$$

Does this expression ring a bell?

$$R_\infty = \left(\frac{1}{4\pi\epsilon_0}\right)^2 \frac{e^4 m_e}{4\pi\hbar^3 c}$$

Bohr found that R_∞ was very close to R_H.

Bohr's model: Emission frequencies

1. The normal state of an atom will be the state with the lowest energy, i.e. $n = 1$ (ground state).

2. In an electric discharge, the atom receives energy due to collisions and the electron makes a transition to a state of higher energy, $n > 1$ (excited state).

3. An excited atom can decay to lower-energy states by emitting photons through allowed transitions. For example, suppose that the electron is excited to the $n = 7$ state and then drops successively through the states $n = 5$ and $n = 3$ to the ground state $n = 1$. Three lines are emitted.

4. In an experiment all possible transitions occur and the complete spectrum is emitted.

Bohr's model: Emission frequencies

Hydrogen (Z=1). Consider the subset of spectral lines which arises from transitions in which $n_f = 2$. Those are

$$\kappa = R_\infty \left(\frac{1}{2^2} - \frac{1}{n_i^2} \right)$$

which coincides with Balmer's formula, provided $R_\infty = R_H$.

The other known series arise from transitions in which the electron goes to a final quantum state n_f. Lyman (n_f =1), Paschen (n_f =3), Brackett (n_f =4), Pfund (n_f =5).

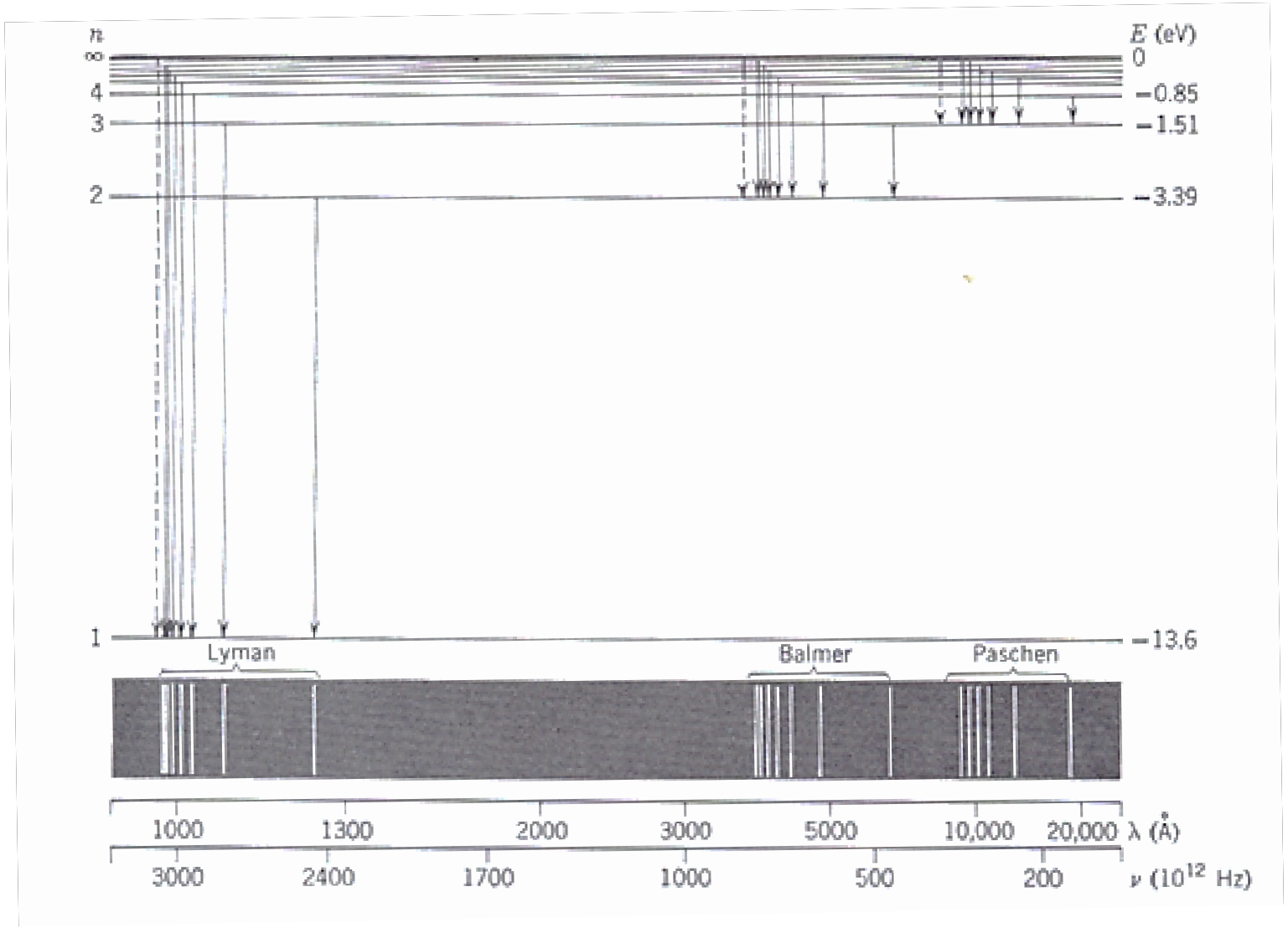
Bohr's model naturally reproduces the known hydrogen spectral series and predicts additional series corresponding to other final quantum numbers.

The model also works for one-electron atoms in which Z ≠ 1

Bohr predicted some spectral lines for Li^{++} y Be^{++}, which subsequently were found in spectral lines from stars.

Absorption spectrum?

Bohr's model: Emission frequencies



Correction for finite nuclear mass

Bohr obtained his formulas by assuming that the nucleus' mass is infinitely large in comparison with the electron's mass ($m/M \rightarrow 0$).

More realistic picture: The nucleus and the electron move about their common center of mass (with same angular velocity ω). ($v_n/r_n = v_e/r_e = \omega$)

$r = r_e + r_n$

Definition center of mass $\Rightarrow$ $Mr_n = m_e r_e$

$$r_e = \left(\frac{M}{M + m_e}\right) r \qquad r_n = \left(\frac{m_e}{M + m_e}\right) r$$

$$\mu = \left(\frac{M m_e}{M + m_e}\right)$$

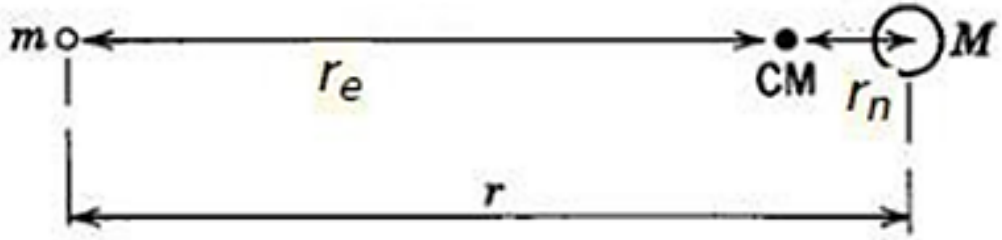


Figure 5–6. The center of mass of an electron-nucleus system.

Correction for finite nuclear mass

Mechanical stability :

$$\frac{1}{4\pi\epsilon_0}\frac{Ze^2}{r^2} = m_e\frac{v_e^2}{r_e} = m_e\omega^2 r_e$$

Second postulate: Total orbital angular momentum of the atom

$$L = Mv_n r_n + m_e v_e r_e = n\hbar \qquad M\omega r_n^2 + m_e\omega r_e^2 = n\hbar$$

$$v_n = \omega r_n \qquad v_e = \omega r_e \qquad \mu\omega r^2 = n\hbar$$

$$\mathrm{r} = 4\pi\epsilon_0\frac{n^2\hbar^2}{\mu Ze^2} \qquad r_\infty = 4\pi\epsilon_0\frac{n^2\hbar^2}{m_e Ze^2}$$

$$\mathrm{v_e} = \frac{\mu}{m_e}\frac{Ze^2}{4\pi\epsilon_0}\frac{1}{n\hbar} \qquad \mathrm{v_{e,\infty}} = \frac{Ze^2}{4\pi\epsilon_0}\frac{1}{n\hbar}$$

n = 1,2,3…

Correction for finite nuclear mass

Potential energy :

$$V = -\frac{1}{4\pi\epsilon_0}\frac{Ze^2}{r}$$

Kinetic energy :

$$K = \frac{1}{2}m_e v_e^2 + \frac{1}{2}Mv_n^2 = \frac{1}{2}\omega^2(m_e r_e^2 + Mr_n^2) = \frac{1}{2}\mu\omega^2 r^2$$

Total energy

$$E_{total} = -\frac{Z^2 e^4 \mu}{2\hbar^2(4\pi\epsilon_0)^2}\frac{1}{n^2} \qquad E_\mu = \frac{\mu}{m_e}E_\infty$$

Now, from $\nu = (E_i - E_f)/h$ we obtain

$$\kappa = R_M Z^2\left(\frac{1}{n_f^2} - \frac{1}{n_i^2}\right) \qquad R_M = \left(\frac{1}{4\pi\epsilon_0}\right)^2 \frac{\mu e^4}{4\pi\hbar^3 c}$$

Notice that $R_M = \frac{\mu}{m_e}R_\infty$

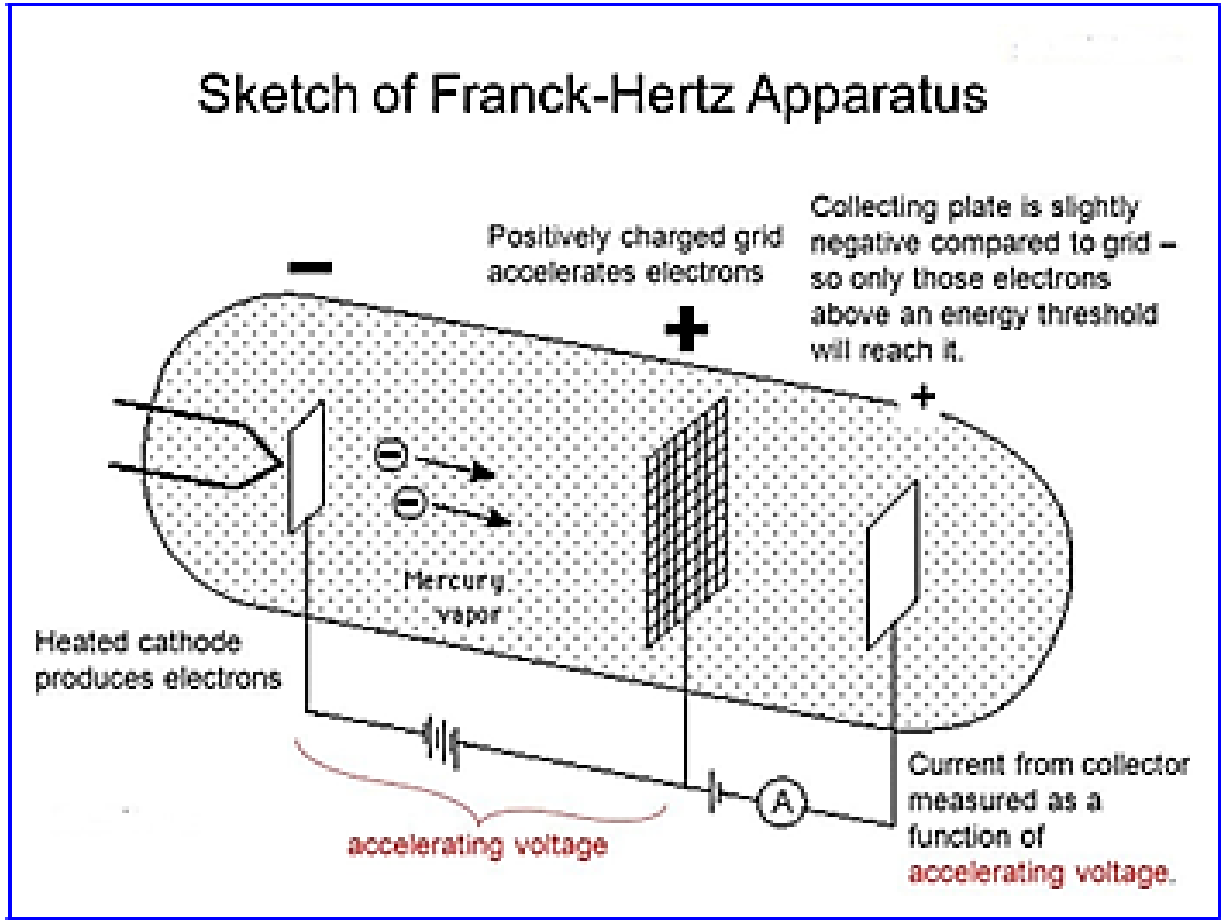
For Hydrogen, $R_M(th) = 10968100\, m^{-1}$, $R_M(exp) = 10967757.6\, m^{-1}$

Atomic energy states

Bohr's model predicts that the total energy of an atomic electron is quantized. It is reasonable to assume that the total energy of each electron in a multielectron atom will also be quantized and therefore the same must apply for the atom's total energy.

Franck and Hertz (1914). They showed that the total internal energy states of any atom are quantized.

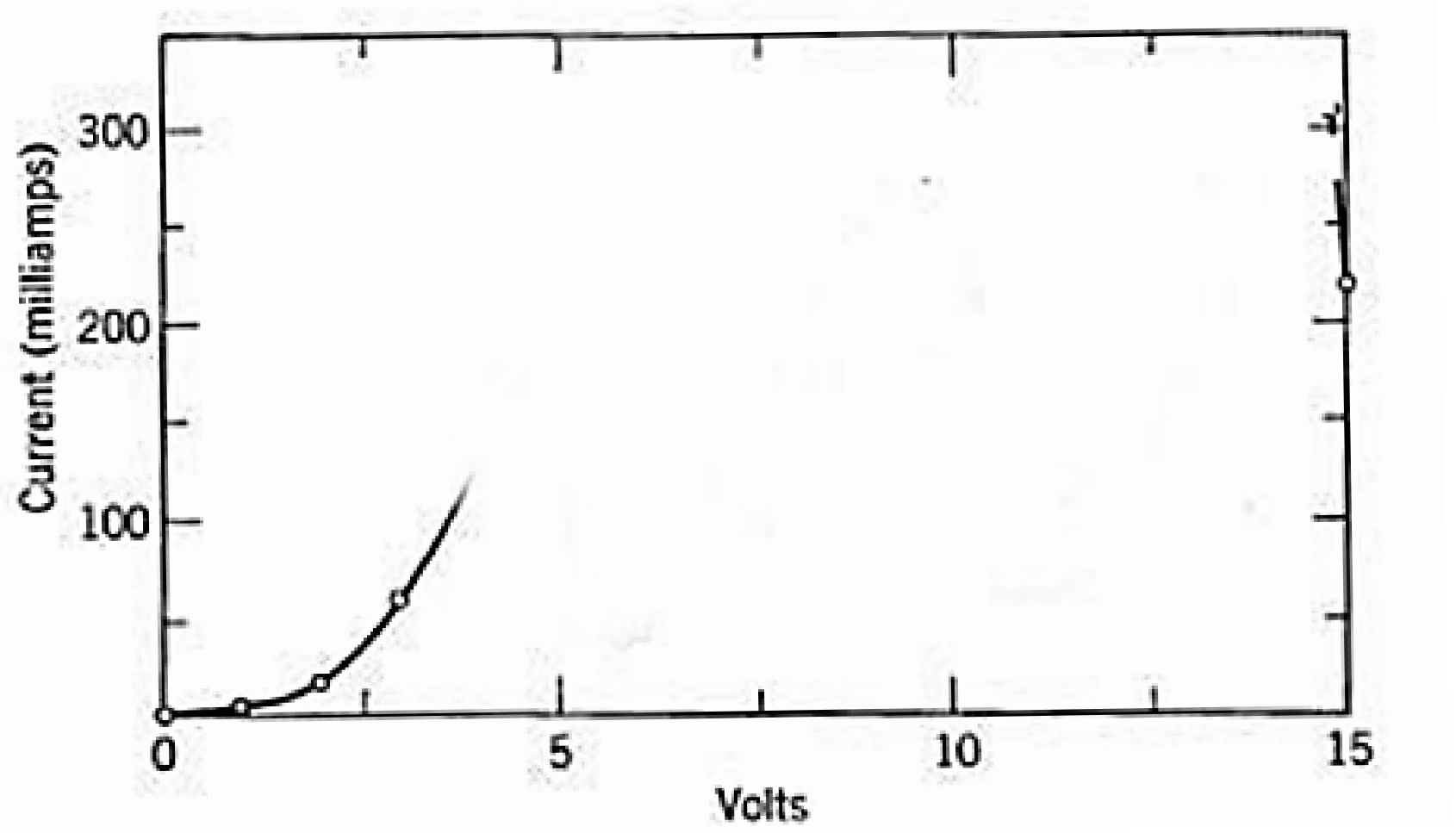
Objective: to measure the electron current reaching the collecting plate as a function of the accelerating voltage V



Atomic energy states

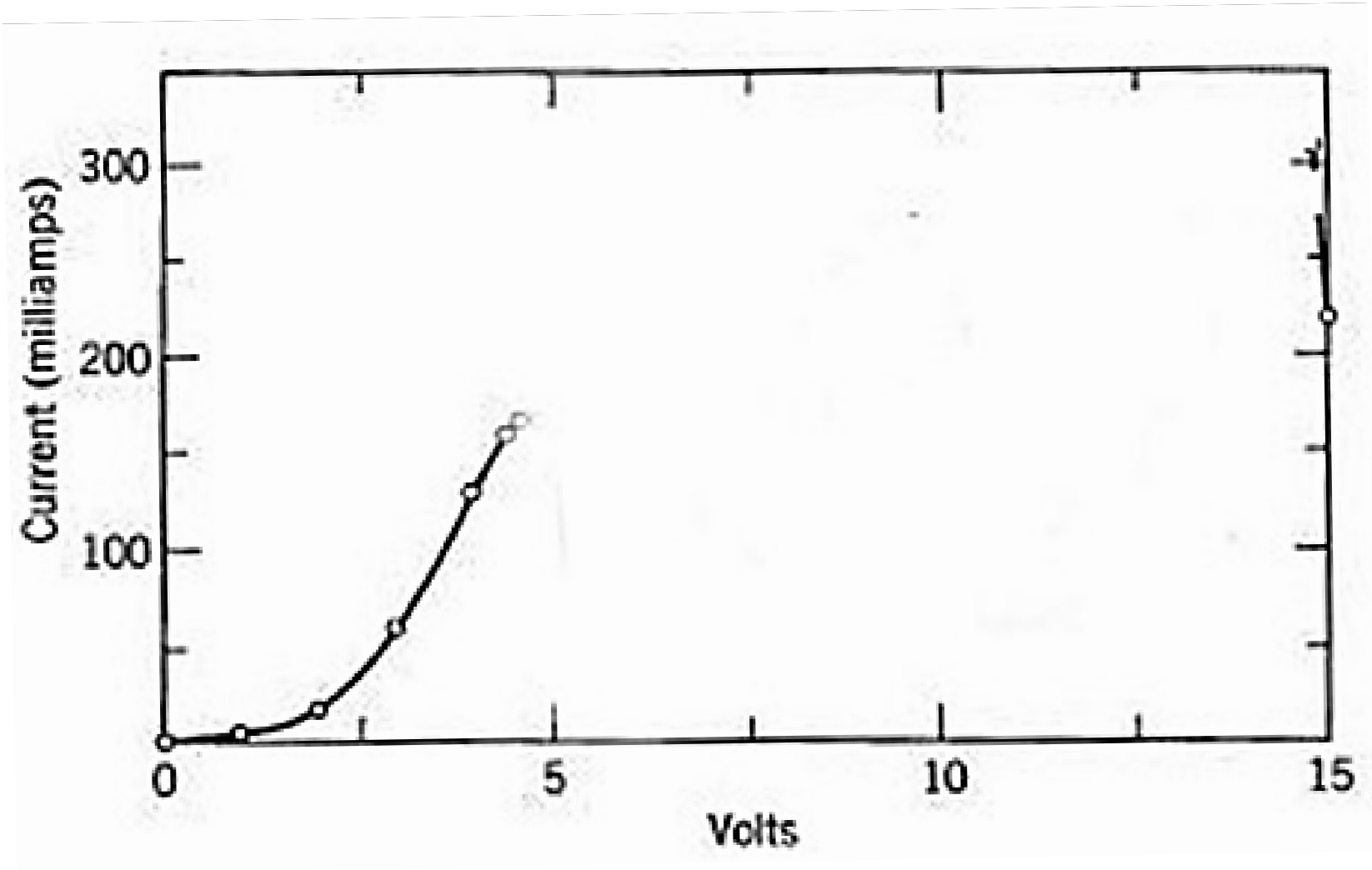
Obtained results

According to classical physics, as the accelerating voltage increases, what should happen to the collector current?



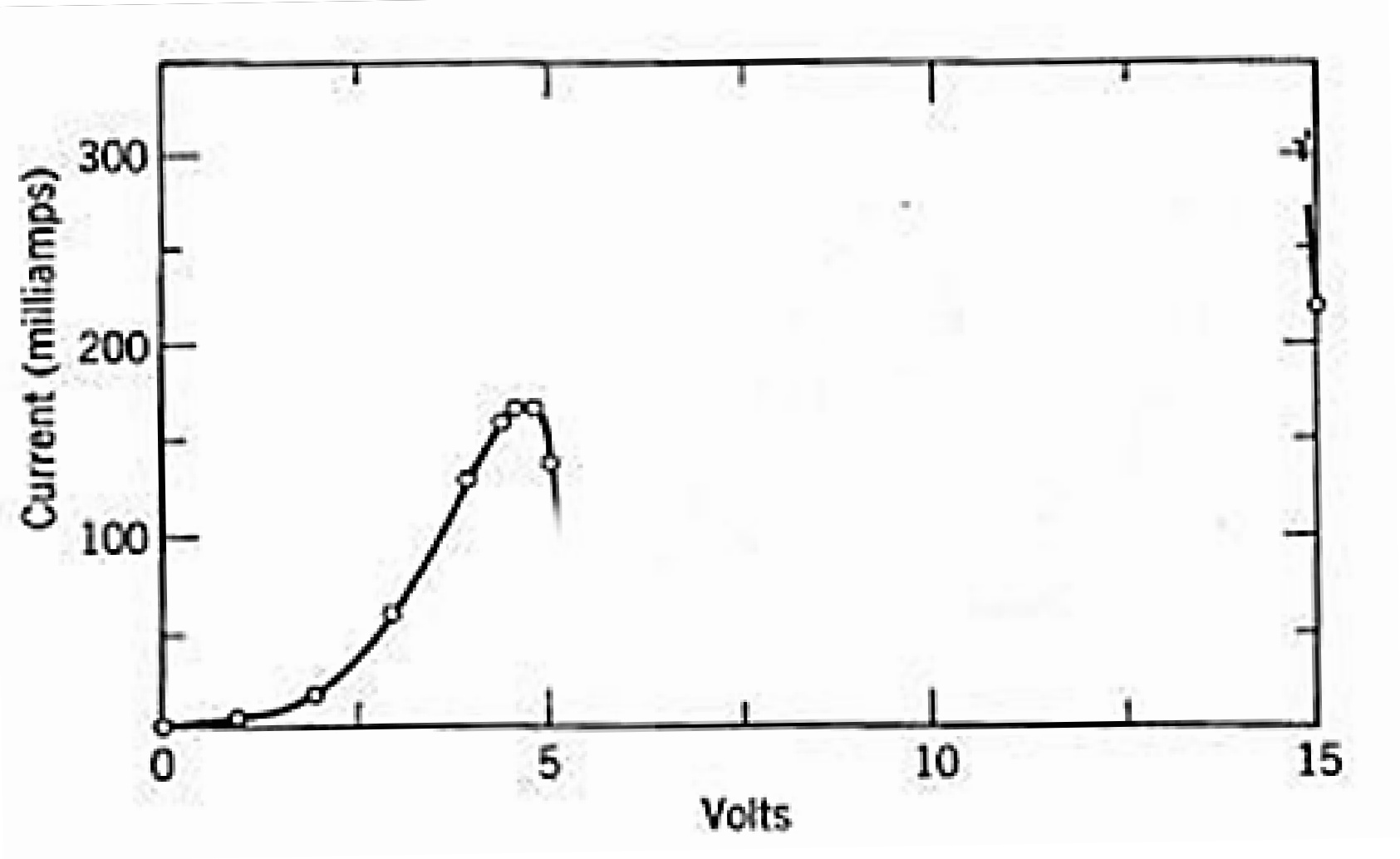
Atomic energy states

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Atomic energy states

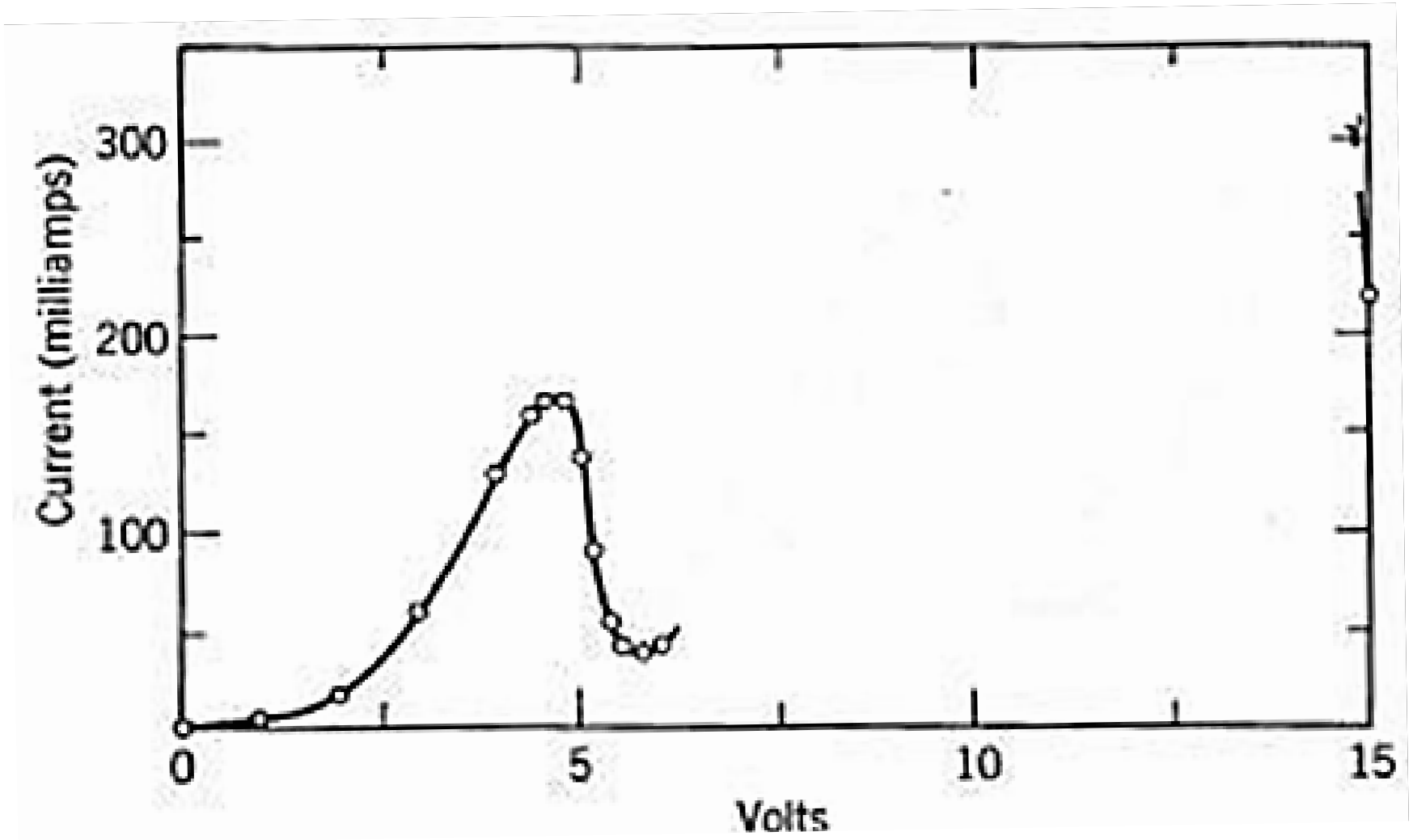
Obtained results



Atomic energy states

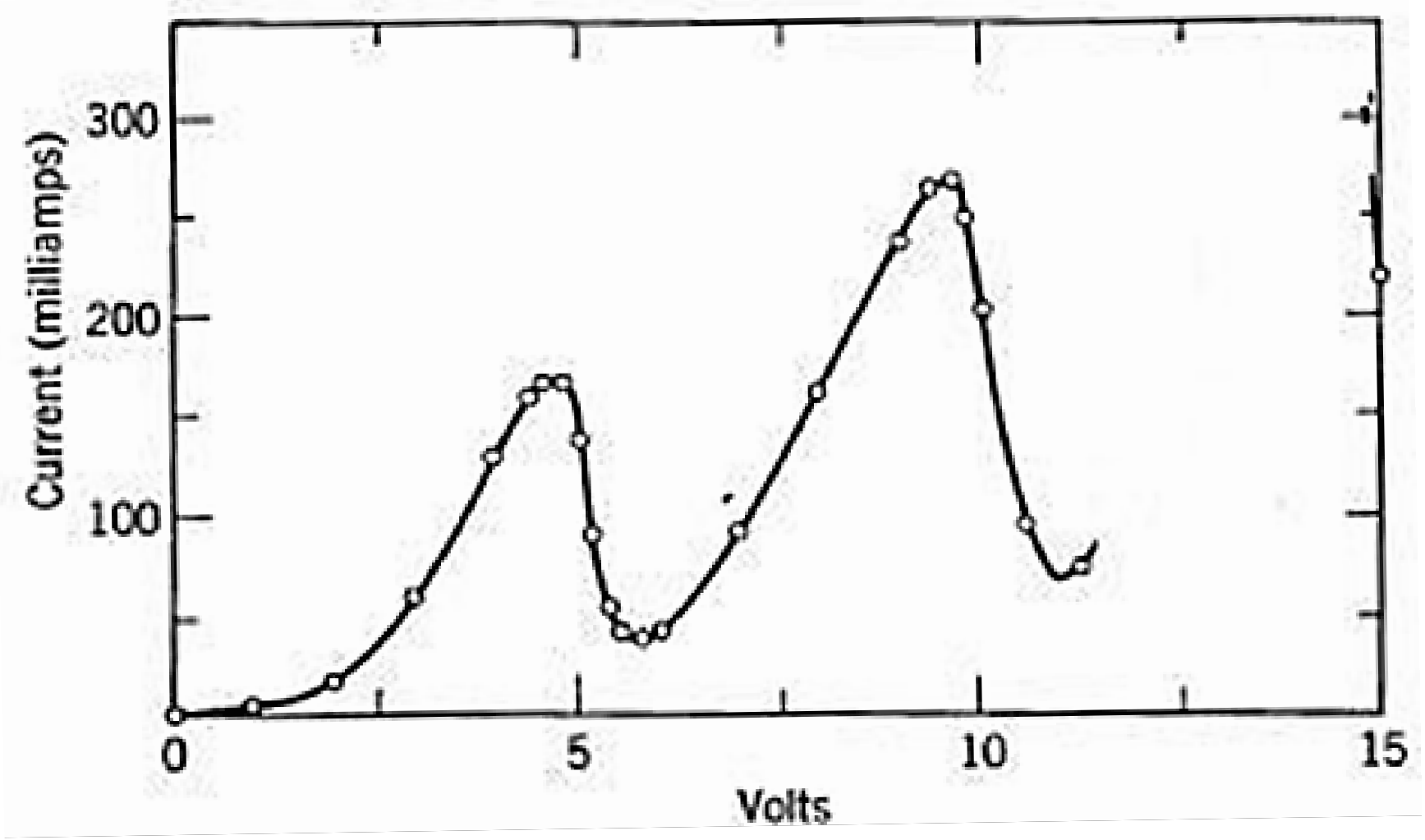
Obtained results

Why does the current suddenly decrease at about 4.9 V?



Atomic energy states

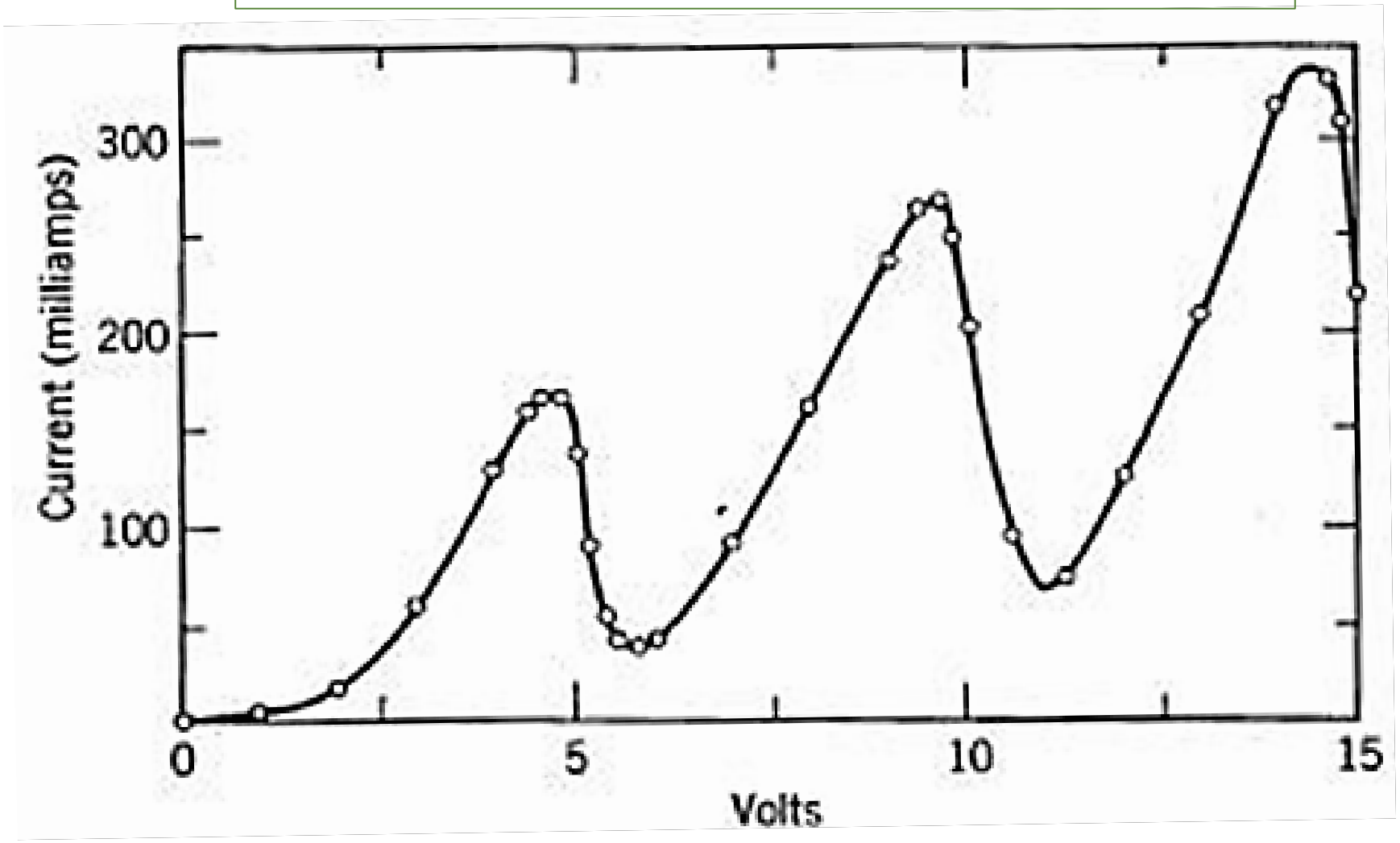
Obtained results



Atomic energy states

Obtained results

Why does the pattern repeat near 9.8 V, 14.7 V, ...?



Atomic energy states

If an electron has a kinetic energy K_e < 4.9 eV, then there is an elastic collision $(\Delta K = \frac{4m}{M} K_e)$

$$e^-(K_e) + Hg(rest) \rightarrow Hg(\Delta K) + e^-(K_e - \Delta K)$$

If an electron has a kinetic energy K_e > 4.9 eV, then there is an inelastic collision $(\Delta K = E_2 - E_1)$

$$e^-(K_e) + Hg(rest) \rightarrow Hg(excited) + e^-(K_e - (E_2 - E_1))$$

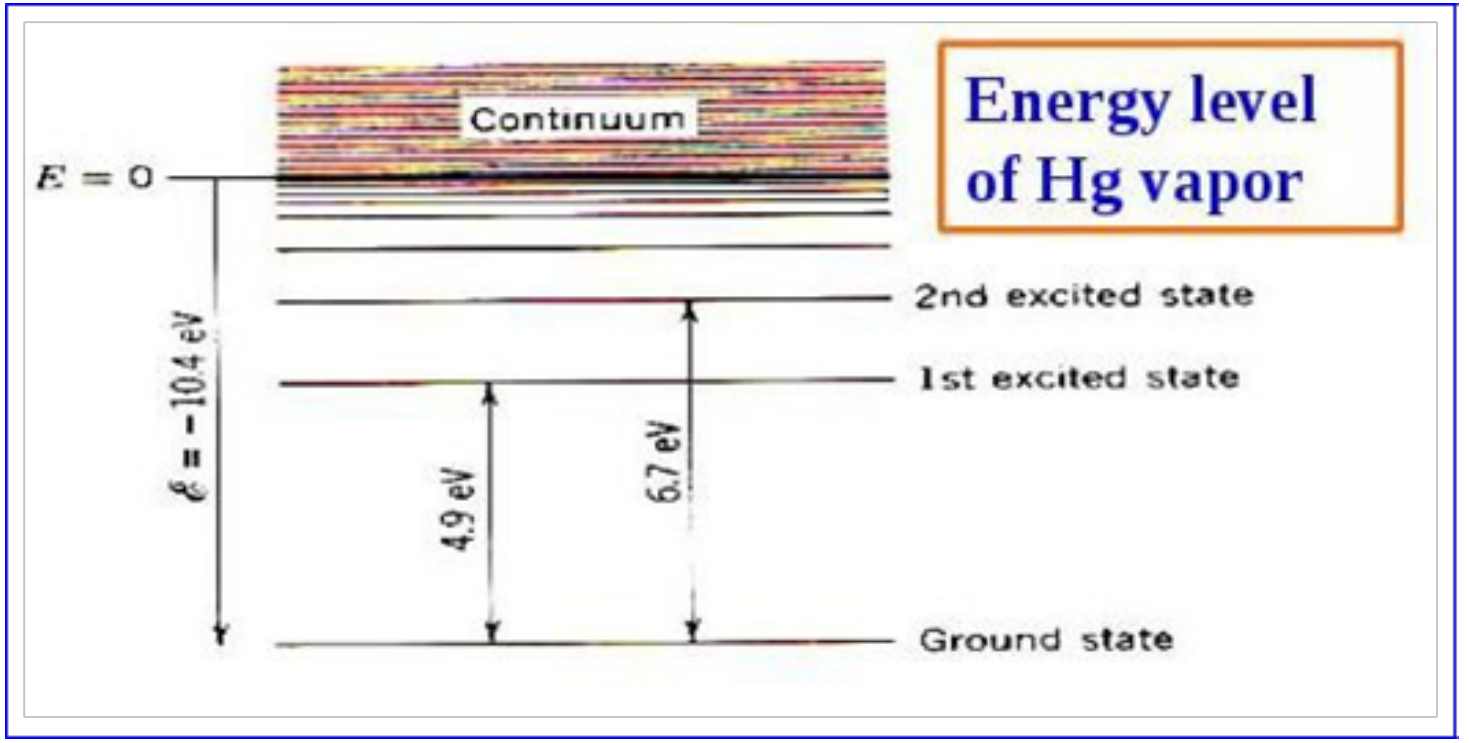
$$Hg(excited) \rightarrow Hg(ground\ state) + h\nu\ (photon\ \lambda = 2536\ \text{Å})$$

Atomic energy states

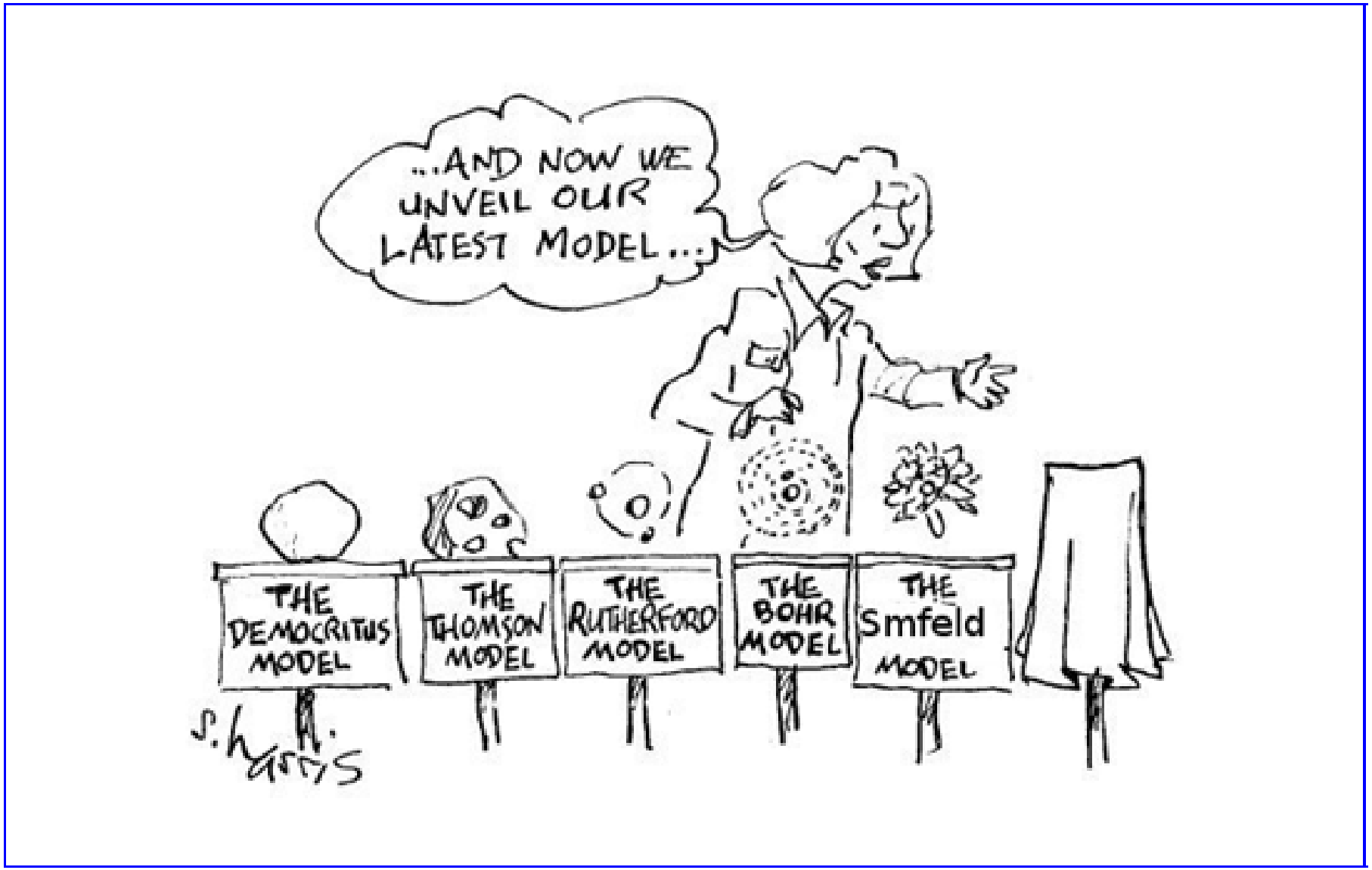
- The total energy of an unbound electron (i.e. free electron with E > 0) is not quantized.

- The electron can be excited to a continuum state if it receives an E ≥ 10.4 eV

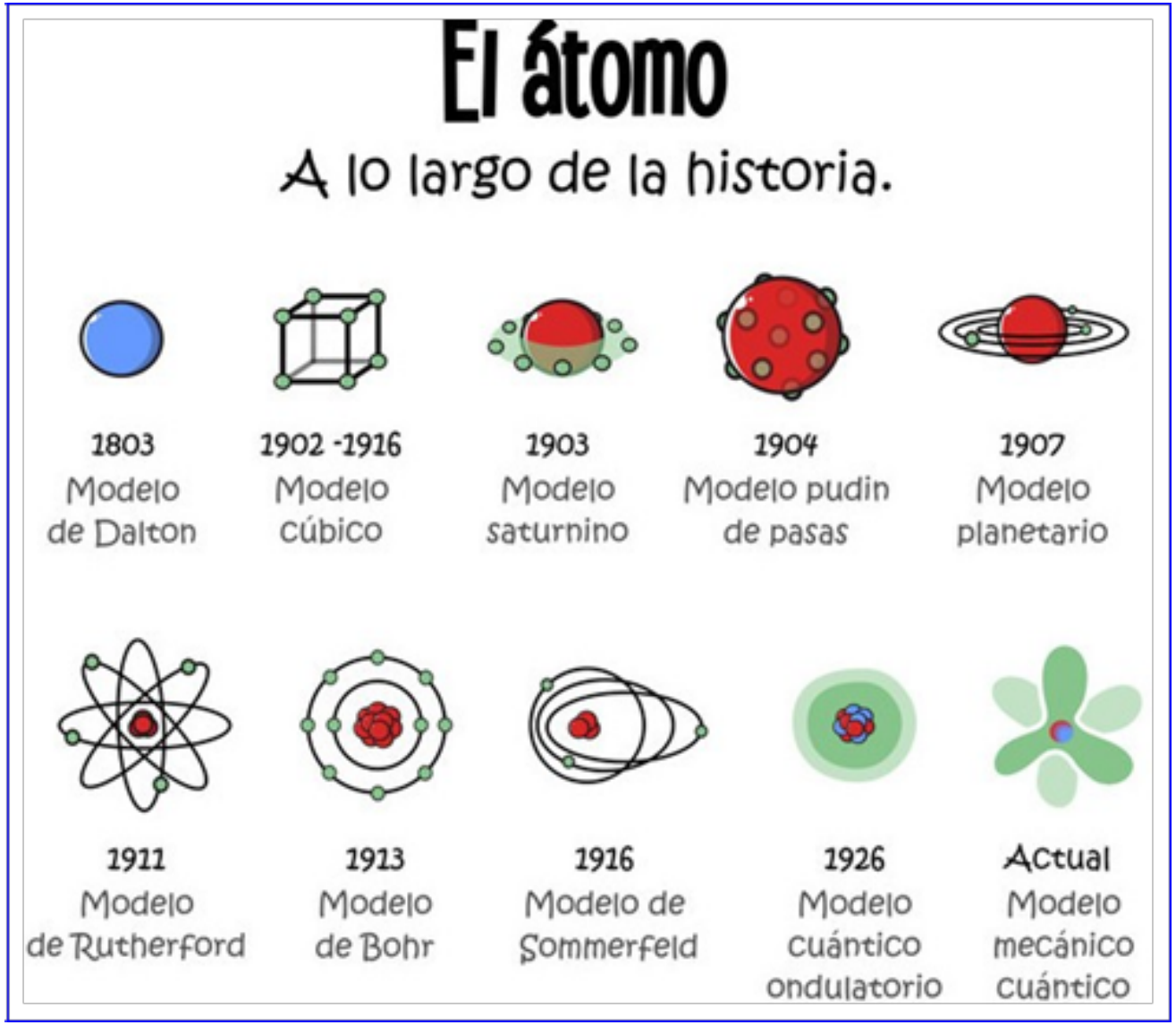
- A Hg^+ ion can capture a free electron into any quantized energy level of the neutral atom. Then radiation is emitted with a υ greater that the series limit.



Endless story



Endless story



The correspondence principle: Bohr

We expect that the quantum theory must correspond to the classical theory in the limit in which the system behaves classically. **Where is the classical limit?**

Bohr (1923). The predictions of the quantum theory for the behavior of any physical system must correspond to those of classical physics in the limit in which the quantum numbers specifying the state of the system become very large.

. Apply the correspondence principle to hydrogen atom radiation in the classical limit.

► The frequency of revolution ν_0 of an electron in a Bohr orbit .. is given by

$$\nu_0 = \frac{v}{2\pi r} = \left(\frac{1}{4\pi\epsilon_0}\right)^2 \frac{me^4}{4\pi\hbar^3}\frac{2}{n^3}$$

According to classical physics the frequency of the light emitted in such a case is equal to ν_0, the frequency of revolution.

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$$\nu = \frac{c}{\lambda} = c\kappa = \left(\frac{1}{4\pi\epsilon_0}\right)^2 \frac{me^4}{4\pi\hbar^3}\left[\frac{1}{n_f^2} - \frac{1}{n_i^2}\right]$$

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But, if this is to agree with ν_0, we must have $n_i - n_f = 1$. To see this, take $n_i - n_f = 1$ and obtain

$$\nu = \left(\frac{1}{4\pi\epsilon_0}\right)^2 \frac{me^4}{4\pi\hbar^3}\left[\frac{1}{(n-1)^2} - \frac{1}{n^2}\right] = \left(\frac{1}{4\pi\epsilon_0}\right)^2 \frac{me^4}{4\pi\hbar^3}\left[\frac{2n-1}{(n-1)^2 n^2}\right]$$

where $n_i = n$ and $n_f = n - 1$. Then as $n \to \infty$ the expression in the square brackets above approaches $2/n^3$ so that $\nu \to \nu_0$ as $n \to \infty$.

The correspondence principle: Bohr

The Correspondence Principle for Hydrogen

n	ν_0	ν	% Difference
5	5.26×10^{13}	7.38×10^{13}	29
10	6.57×10^{12}	7.72×10^{12}	14
100	6.578×10^{9}	6.677×10^{9}	1.5
1,000	6.5779×10^{6}	6.5878×10^{6}	0.15
10,000	6.5779×10^{3}	6.5789×10^{3}	0.015

A critique of the old quantum theory

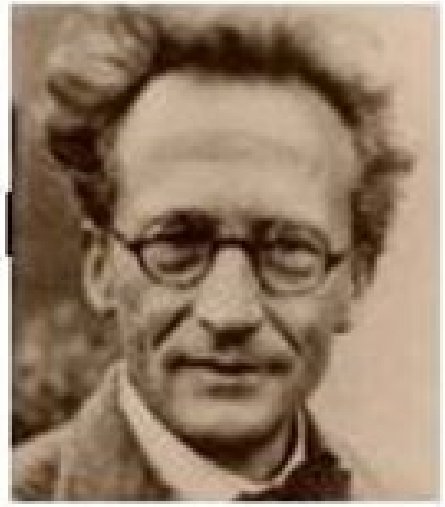
- It lacks internal consistency because it mixes classical laws and quantal postulates. According to W. Bragg: ”The classical theory is used on Mondays, Wednesdays and Fridays whereas the quantum theory is used on Tuesdays, Thursdays and Saturdays.”
- The atomic stationary states are postulated but the theory is unable to justify its existence.
- The theory only tells us how to treat systems which are periodic, by using the Wilson-Sommerfeld quantization rules, but there are many systems which are not periodic.
- The theory is successful for one-electron atoms. But it fails even for the neutral He atom, which contains only two electrons.
- It does not provide any explanation for the double nature of matter and radiation.
- It can not explain the anomalous Zeeman effect.

Next step

To solve this situation, a new conceptual perspective was needed and this was initiated by Schrödinger and Heisenberg in 1925



Werner Heisenberg



Schrödinger

The Schrodinger theory can be considered as a generalization of the de Broglie postulate.